GEOG 172

# Think Geographically, Measure Locally

Lecture 7-1

# LOCAL MORAN'S I

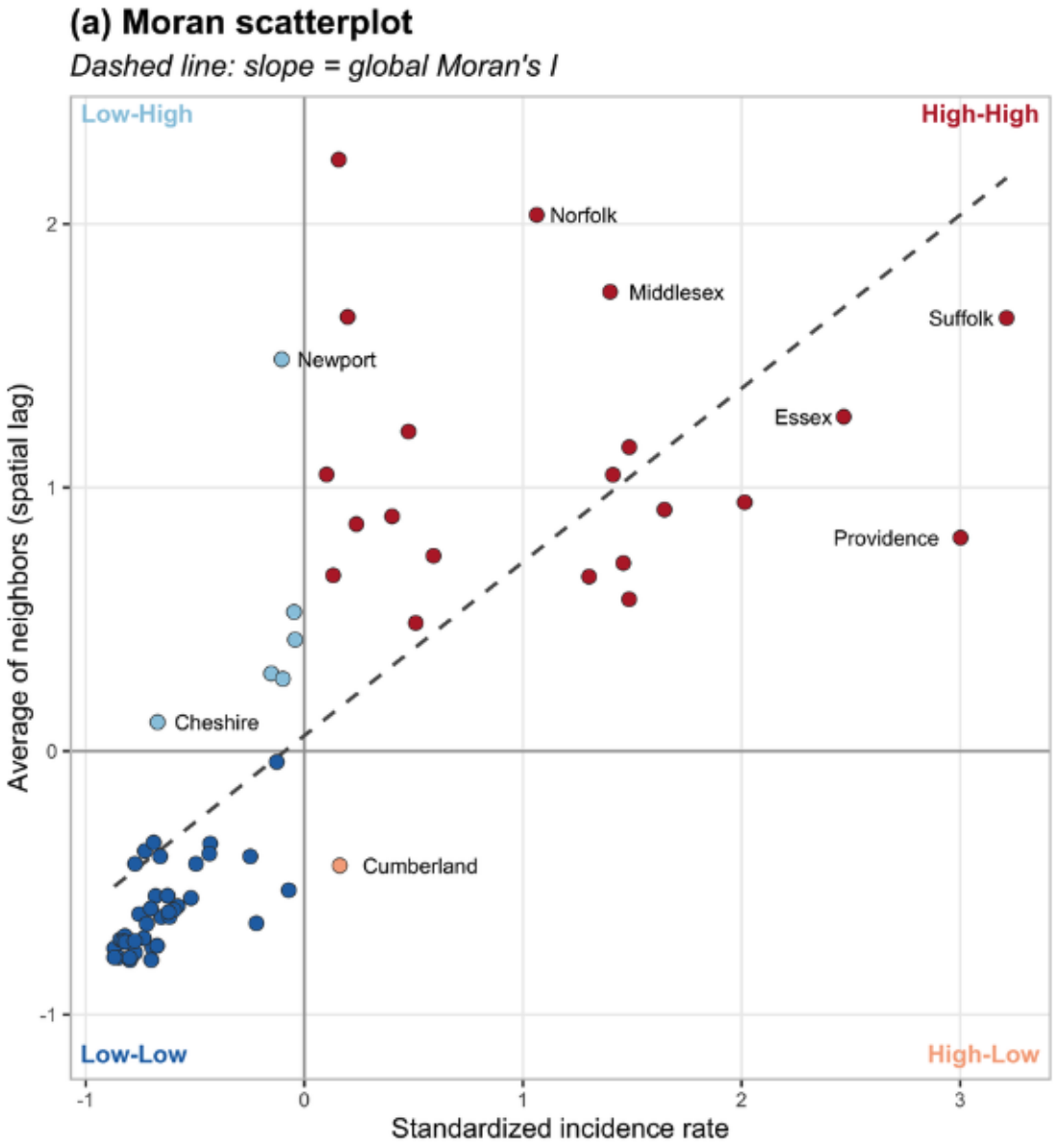
$$I_i(d) = (x_i - \bar{x}) \sum_{j=1}^{n} w_{ij}(x_j - \bar{x}) = z_i \sum_{j=1}^{n} w_{ij} z_j$$

- Product of the deviation from the mean at i, and the deviations from the mean surrounding i
- **Positive**: similar values side by side (high–high or low–low), a *cluster*
- **Negative:** dissimilar values side by side (high next to low), a spatial outlier

$I_i$ measures homogeneity versus diversity, not whether values are high or low

# MORAN SCATTERPLOT

- x-axis: a location's standardized value.
- y-axis: the average of its neighbors (the spatial lag)
- Lines at the mean divide the plot into four quadrants
- Slope of the fitted line = global Moran's I



# IMPLEMENTATION IN R

```R
localmoran(x, listw, zero.policy = attr(listw, "zero.policy"), na.action =
na.fail, conditional = TRUE, alternative = "two.sided", mlvar = TRUE,spChk =
NULL, adjust.x = FALSE)
```

- x — A numeric vector of attribute values, one per location, in the same order as the neighbors list. The function centers the values $(z_i = x_i - \bar{x})$.
- listw — The spatial weights object (built with nb2listw()). This is where the analyst's definition of "local" enters the calculation.
- conditional —The default, TRUE, uses conditional randomization (holding x_i fixed), matching the logic of the permutation approach discussed earlier; FALSE uses total randomization.
- alternative — The direction of the hypothesis test: "two.sided" (default, hot *or* cold spots), "greater" (hot spots only), or "less" (cold spots only).

# IMPLEMENTATION IN R

```R
localmoran_perm(x, listw, nsim = 499, zero.policy = attr(listw, "zero.policy"),
                na.action = na.fail, alternative = "two.sided", mlvar = TRUE,
                spChk = NULL, adjust.x = FALSE, sample_Ei = TRUE, iseed = NULL,
                no_repeat_in_row = FALSE)
```

- **nsim** — The number of permutations (default 499).
- **sample_Ei** — If TRUE (default), the expected value of I_i is taken from the mean of the permuted values; if FALSE, the analytical expected value is used and only the variance comes from the simulation.
- **iseed** — Sets the random seed so the permutations (and the p-values) are the same each time the code is run. Defaults to NULL (a different result each run).
- **no_repeat_in_row** — If TRUE, each location's neighbors are drawn without replacement within a permutation, so the same value cannot appear twice among one location's neighbors. Defaults to FALSE.

# IMPLEMENTATION IN R

R

```
library(spdep)

# Construct the weights matrix
nb <- poly2nb(tracts, queen = TRUE)                    # queen contiguity neighbors
lw <- nb2listw(nb, style = "W", zero.policy = TRUE)  # row-standardized weights

# Calculate Local Moran's I using defaults
lisa <- localmoran(tracts$income, lw)

# Inspect elements returned by localmoran
tracts$Ii   <- lisa[, "Ii"]                            # local Moran's I
tracts$Z_Ii <- lisa[, "Z.Ii"]                          # z-score
tracts$p_Ii <- lisa[, 5]                               # p-value, column "Pr(z !=
E(Ii))"
tracts$quad <- attr(lisa, "quadr")$mean                # High-High, Low-Low, Low-
High, High-Low
```

# Significance Testing

*Area Relationships Revisited*

## P1: TRUSTING THE NORMAL DISTRIBUTION

In practice, it is difficult to justify the assumption of normality for local statistics

- Local statistics are built using only a few neighbors
- Small neighborhoods, edge locations, and skewed data are all possible violations of normality
- Making local regions bigger helps, but then they're less local

## CONDITIONAL PERMUTATION

Like Monte Carlo simulation, we can use re-shuffling as a means of simulating CSR

- Hold location i's value fixed
- Shuffle all the other values among its neighbors
- Recompute the statistic, and repeat 999 or 9,999 times
- Rank the observed value among the shuffled ones to get the pseudo *p*-value

## P2: MULTIPLE TESTING

When carrying out many local tests, the multiple testing being done needs to be accounted for. For example, if we calculate local spatial statistics for a map with 100 sub-regions and used the 5-percent significance threshold, we would expect 05 false positive signals under CSR.

- Neighboring locations share information, so tests are not independent
- Permutation does not solve this problem

# P2: MUTLTIPLE TESTING

When carrying out many local tests, the multiple testing being done needs to be accounted for.

## (A) Ignore the Problem

Not the best choice because the problem will still exist and your inferences could well be wrong, or right, but you won't know

## P2: MUTLTIPLE TESTING

When carrying out many local tests, the multiple testing being done needs to be accounted for.

(A) Ignore the Problem (Bad Idea)

(B) Bonferroni Adjustment

Divide the critical value of the test statistics by the number of tests.

$\alpha = 0.05/100 = 0.0005 \rightarrow Zcrit = 3.29$

## P2: MUTLTIPLE TESTING

When carrying out many local tests, the multiple testing being done needs to be accounted for.

(A) Ignore the Problem (Bad Idea)

(B) Bonferroni Adjustment (Too Conservative)

(C) Sidak Adjustment

Sidak has the nice property of producing an error rate of α when tests are independent and Ho are true

$$\alpha_{Sidak} = 1 - (1 - \alpha)^{\frac{1}{m}}$$

## P2: MUTLTIPLE TESTING

When carrying out many local tests, the multiple testing being done needs to be accounted for.

(A) Ignore the Problem (Bad Idea)

(B) Bonferroni Adjustment (Too Conservative)

(C) Sidak Adjustment (Less Conservative)

(D) False Discovery Rate (FDR)

- Estimates the number of false positives based on the distribution of *p*-values and the number of tests.
- Significant p-values are ranked from smallest to largest, and some of largest are then removed based on the FDR
- The FDR falls between doing nothing and the Bonferroni or Sidak (family-wise error rate) corrections.

# Local Statistics

*From concept to computer*

# TUTORIAL

Where, beneath the 2024 sea of red, are the real geographic concentrations of partisan support and of political change, and which of them are strong enough that they are unlikely to be chance?

- A county map can make a pattern look obvious, and a global statistic can confirm that the vote is clustered.
- Neither tells us which places form those clusters, or whether a cluster of a few counties is more than the result of testing thousands of counties at once.

In this tutorial you will move from the national picture to the county, find where support and change are concentrated, and decide which of those concentrations you would be willing to report.